DE.AUTO

CHAPTER 14

MAINTENANCE & CARE

THE WORLD OF CAR MODIFICATIONS MADE SIMPLE

Cars explained in plain, friendly English

MEMBER EXCLUSIVE · DE.AUTO ·

POWERED BY DE.AUTO

DE.AUTO — Because knowledge is power, and cars are passion.

CAR MODIFICATIONS GUIDE MUST-KNOW TERMS

Car modifications, explained

Chapter Fourteen: Maintenance and Care

“A GREAT CAR IS A WELL-LOVED CAR.”

— DE.AUTO

Important notice

This guide is **entertainment only**. It is not educational or scientific content, and it is not a how-to manual. DE.AUTO is not an accredited scientific, engineering or technical authority. Everything here is simplified for fun and general interest, and it can differ from one car to another.

You must **never** try any modification, drive at high speed, or do any kind of stunt or dangerous driving on public roads. Safety and respecting traffic laws are everyone's own responsibility.

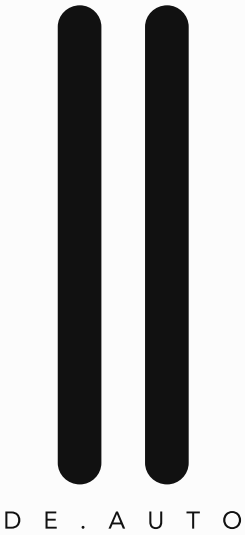
The information here is **general and approximate**, and much of it describes only the typical case. Laws, regulations and specifications differ by country and model, so always check the owner's manual and the official authorities.

Vehicle modifications are subject to local laws. Any unauthorized modification can lead to legal liability, fines or having your vehicle impounded.

DE.AUTO is not responsible for anyone who applies the information in this guide in real life.

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For any decision about your car, go to certified professionals and the manufacturer's manual, and only try something if it complies with official regulations and legal requirements. 🚀



DE.AUTO is more than a name. It's an identity that brings the passion for cars together with knowledge and progress.

At DE.AUTO we make fresh content for car lovers, mixing fun and know-how from the world of modifications and high performance, in a simple style that's easy to follow. Our goal is to help you understand your own car better and make smarter calls on modifications and maintenance.

We walk you through the key terms and parts of the modification world, so you can chat with the enthusiasts and follow every detail without feeling like an outsider.

Whether you're a beginner or a pro, you'll find terms, basics and fun facts here that help you get into cars in an organized, confident way.

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READY?

You spent serious money on modifications… And when did you last check the oil level? 😬

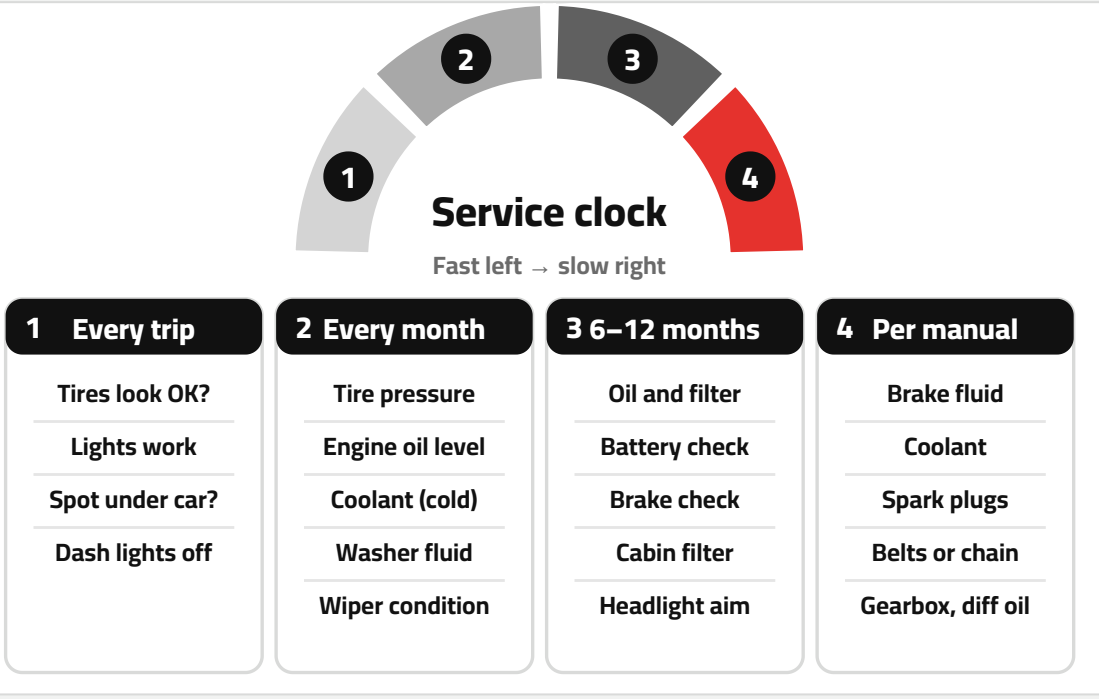
By the end of this chapter you'll know:

- ✓ Why maintenance is the foundation that everything in the series is built on
- ✓ "The maintenance clock": what to check and how often
- ✓ Oil: how to read 5W-30 and how often to change it in general
- ✓ Coolant, brake fluid and gearbox oil in Gulf heat
- ✓ Warning lights: red means stop safely, amber means check soon
- ✓ Paint and interior care, and the difference between wax, ceramic and PPF
- ✓ A used-car buying checklist and how to choose a workshop you trust



Maintenance: the foundation everything stands on 🧱

Every chapter you have met in this series, from the engine and turbo to suspension and brakes, works on one condition: **that it is healthy to begin with**. Any modification on top of a neglected base is like a beautiful house built on soft ground. 🏠



Illustrative drawing: the intervals are general and vary from car to car; the first reference is always your owner's manual

✅ **Golden rule:** These intervals are **general** and for illustration. Always follow **your car's manual (Owner's Manual)**, because the manufacturer knows its own car best, and Gulf conditions (heat, dust, traffic) can shorten the intervals.

Pre-trip checklist: just two minutes ⏱️

1. Walk around the car
Tires inflated? Any nails? Any oil or water spot underneath?

2. Lights
Front, rear, brake and indicators. Ask someone to help or use a reflection in glass.

3. Glass and wipers
Clean? Washer fluid there? Blades not dried out?

4. Inside
Warning lights go out after start-up, enough fuel, A/C working, seats and mirrors adjusted.

5. Emergency
Spare tire and tools or a tire sealant kit, a warning triangle, a phone charger, and water. In summer, water is not a luxury! 💧

6. Around you
Make sure there is no child or object behind the car before you move.

🛢️ Engine oil: the car's blood

Oil doesn't just "lubricate". It does five jobs at once, and if any one of them fails, the engine suffers. And it is the cheapest thing you can do for your car. 💰

Reduces friction
Forms a film between the metal parts so they don't rub.

Cools
Carries away some of the heat from the piston and from parts the coolant doesn't reach.

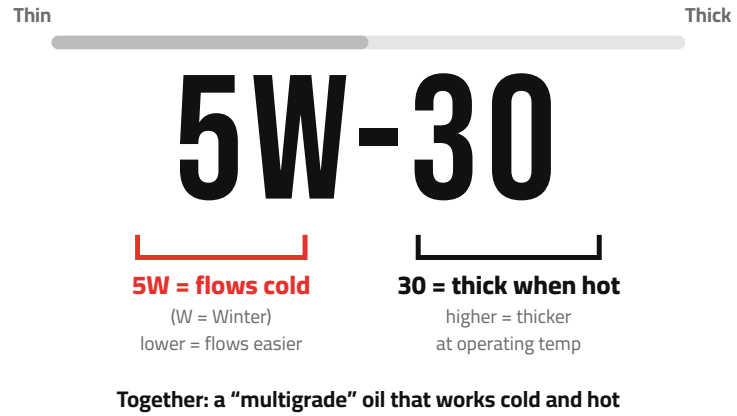
Cleans
Picks up impurities and deposits and carries them to the filter.

Seals
Helps the piston rings seal better.

Protects from rust
Coats surfaces against corrosion and moisture.

No oil = disaster
An engine can't survive minutes without enough lubrication.

How do you read the viscosity number? 🔍



The number before W is for cold, and the number after is for heat. Which grade to use? The manual decides

The three types of oil

Mineral
Simplest and cheapest; intervals are usually shorter.

Semi-synthetic
A blend of mineral and synthetic.

Full synthetic
Higher stability with heat and usually better protection.

💡 **Careful:** Oil cans carry codes such as API and ACEA and manufacturer approvals. These are the "specifications", and viscosity is only part of them. What you need is exactly what your manual asks for, not just the same number.

When do you change the oil? And what about the filter? 🔧

There is no magic number for all cars. The interval depends on the engine, the oil type and how you use the car. This table is **general**, for illustration only:

Condition	Direction	Note
Mineral oil	Shorter interval	Usually the shortest intervals; the manual decides
Full synthetic oil	Longer interval	Many manuals allow a longer interval, but not always
Severe use	Shorten the interval	Short trips, heavy stop-start traffic, dust, high heat, towing
Modified car	Shorten it more	More heat and pressure on the oil; details are on the modifications page

✕ MYTH
"You must change the oil every 1,000 km to protect the engine."

✓ FACT
No ordinary engine needs that frequency. Changing too early just wastes your money. Follow the manufacturer's manual and shorten the interval only if your use is severe.

2 MIN minutes is enough to check the oil level with the dipstick. The easiest and cheapest check, and it protects your engine from big surprises.

The dipstick check 🧪

1. Park on level ground, switch off the engine and wait a few minutes.
2. Pull out the dipstick, wipe it with a clean cloth, and push it all the way back in.
3. Pull it out and check: the oil must be between the MIN and MAX.
4. Look at the colour and feel: darkening over time is normal, but if you see it **milky**, or with a metallic shimmer or a smell of petrol, stop and have a specialist check it.
5. Never go above MAX. Extra oil is not better, and it does harm.

Filters: the engine's guards 🛡️

Oil filter
Traps impurities. Usually changed at every oil change.

Air filter
Keeps dust out of the engine. In the Gulf, check it more often.

Fuel filter
Protects the injectors and the pump. Many modern cars put it inside the fuel tank.

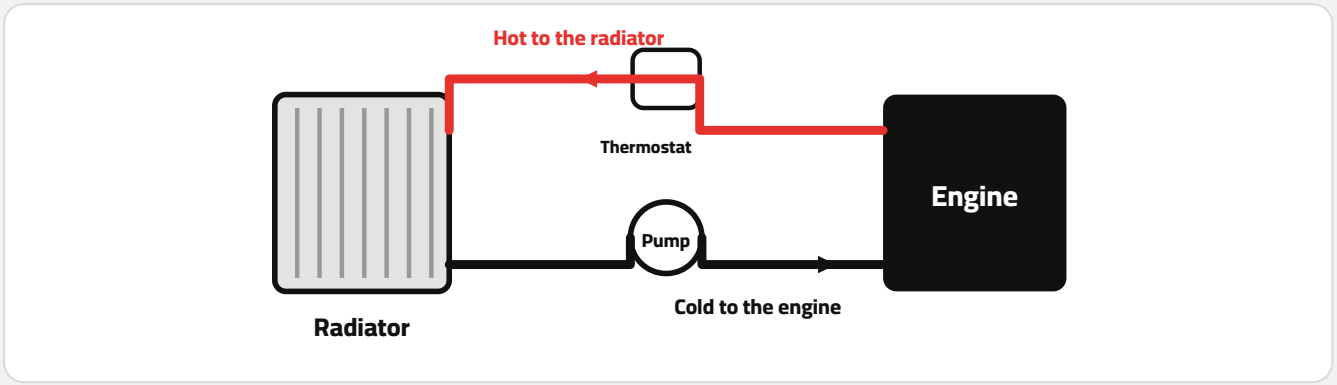
Cabin filter
Cleans the cabin air; if it clogs, ventilation weakens and a smell appears.

🎯 **Challenge 1** **What does the number 5 mean in 5W-30?**

A) Viscosity when hot | B) Fluidity when cold | C) Number of kilometres

🌡️ Fluids: coolant, brake fluid, gearbox oil

Coolant in Gulf heat



A simplified cooling loop. Red lines are hot coolant, black lines are cold

The mix: Coolant is mixed with water as the manufacturer recommends. Water alone boils faster and causes rust, and undiluted coolant isn't always better.

Don't mix types: Colours don't indicate the type. Use the specification your manual states.

Check it cold: the reservoir level should sit between the two marks. And don't open the radiator cap while the engine is hot, because the system is under pressure and can spray scalding liquid on you. ⚠️

The Gulf: Any repeated loss, temperature rising in traffic, or a fan that doesn't work means get it checked before the temperature climbs further.

Brake fluid

2 YR years is a common interval in many manufacturers' recommendations for changing brake fluid. Your manual sets the real figure for your car.

The fluid in your brakes (Brake Fluid, with specifications such as DOT 3, DOT 4 and DOT 5.1) absorbs moisture from the air over time, and that lowers its boiling point. If it boils in hot brakes, you feel a soft pedal. Changing it is a job for a technician, not trial and error.

> 💡 **Heads-up:** The DOT 5 type is silicone-based and doesn't mix with the other types. Read what is printed on the reservoir cap, and use only what is specified.

Gearbox and differential oil

Transmission fluid (ATF for automatics, gear oil for manuals) and differential oil work under pressure and heat. Some manufacturers say "lifetime", but many technicians recommend changing it periodically. Each transmission type (CVT, DCT, and others) needs its own fluid, and any mistake is costly. **Check the manual or the dealer.** And remember Chapters 5 and 6.

🎯 Challenge 2 **What is the safe way to open the radiator cap?**

A) Open it while the engine is running

B) Open it after it has cooled down completely

C) Spray it with cold water

⚙️ Belts, spark plugs, filters and battery

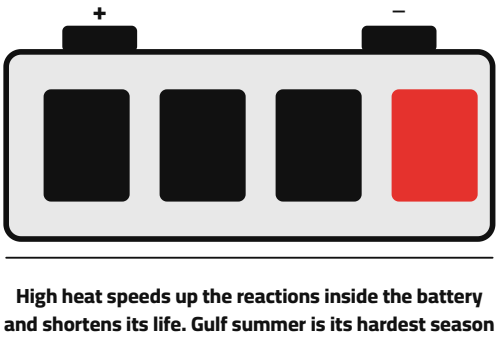
Accessory belts
Drive the alternator, the power-steering pump and the A/C. Check for cracks, glazing and noise (squealing at start-up).

Timing belt or chain
Links the crankshaft to the camshafts. A belt has a fixed life in the manual, and a chain makes a noise when it wears. If it breaks, the engine is badly damaged.

Spark plugs
Copper, platinum or iridium. The interval varies a lot with the type and the car. Modified and turbo cars usually need earlier checks.

Air filter
In dust it gets dirty quickly. Don't blow it out with strong compressed air, and don't add extra oil to it, as that can affect the sensors.

The battery in hot weather 🔋



A simplified drawing of a car battery

Life: In the Gulf it is usually shorter than in cold regions. A weak battery lets you down the moment you need it.

Testing: Ask for a battery test before summer. The test is usually quick and simple.

Terminals: If you see a white or green crust, have a technician clean it carefully. And disconnecting the battery in modern cars affects the systems, so ask first.

Signs of weakness: The starter turns slowly, the lights dim at start-up, and the battery light comes on.

> 💡 **Reminder:** If your car sits for weeks, the battery drains. A small maintenance charger keeps it topped up (see the storage section).

🛞 Tires, suspension and wipers: the quick check

Chapters 8 and 10 explained the details. Here we summarise the “two-minute check” you do yourself:

30 DAY days is a reasonable interval for checking tire pressure when cold. Heat raises the pressure, so never let air out when they are hot.

Pressure

The figure is on the driver’s door sticker or in the manual, not the one printed on the tire sidewall. Check it when the tire is cold.

Tread depth

The wear indicator TWI sits in the grooves. If the tread reaches it, replace the tire. In the Gulf, a worn tire is more dangerous in sudden rain.

Age

The DOT code includes the week and year it was made. An old tire hardens and cracks even if it looks new, so know its age.

Uneven wear

Wear on the inner or outer edge or the centre points to wrong pressure or an alignment and balance problem.

When should you check alignment and balance? 📐

The car pulls to one side while you drive straight.

The steering wheel is crooked while you drive straight.

After a hard hit on a pothole or kerb, or after a suspension change (Chapter 8).

Uneven wear on the tires.

Quick suspension checks 🔩

1. **Bounce test:** Press down hard on a corner of the car and let go. If it bounces more than once, the shock absorber may be weak.
2. **Leaks:** Look at the shock absorbers from the side. A sticky oily trace means a leak.
3. **Noises:** Knocking over bumps or when turning needs a check.
4. **Rubber boots and bushings:** Visible cracks mean it is time to replace them.

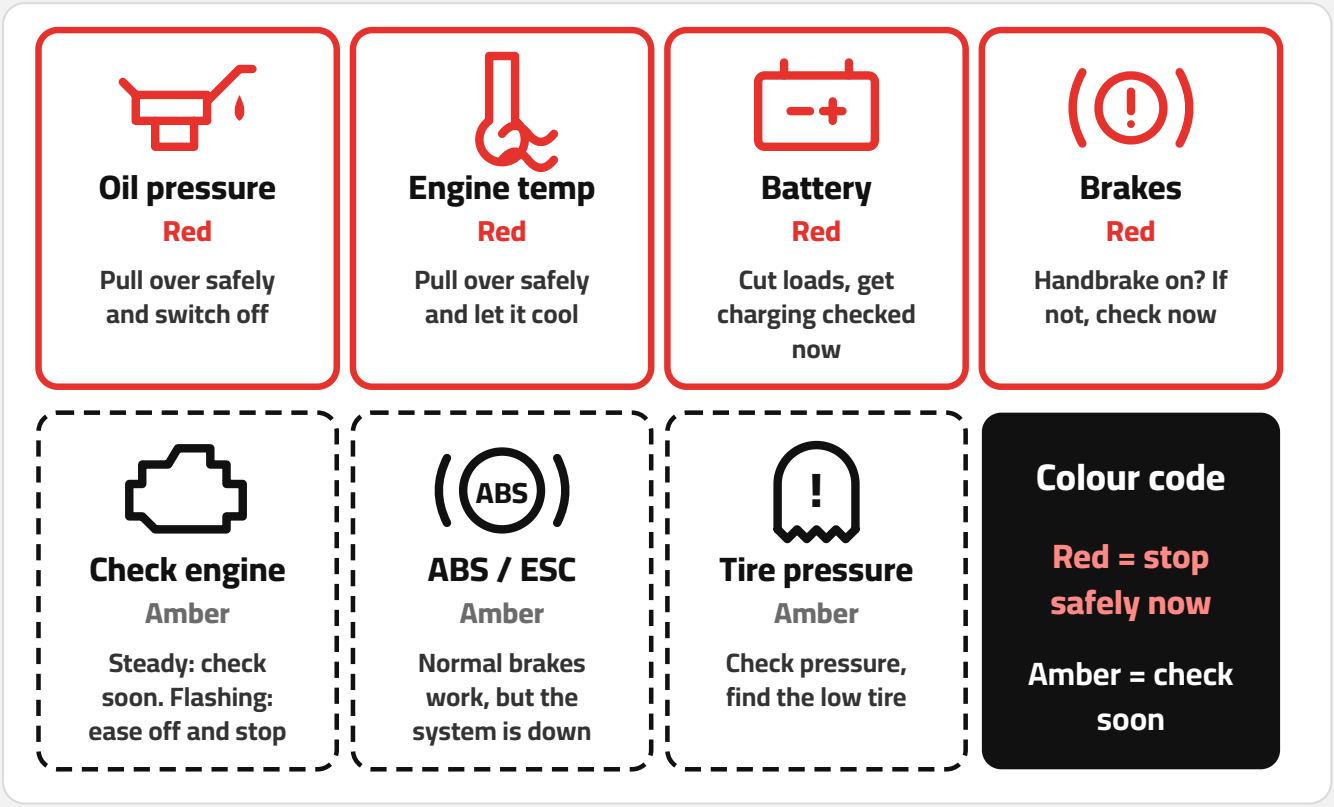
Wipers and lights 💡

The Gulf sun dries wiper blades quickly, so they start to streak the glass and blur your vision. Replacing them is very cheap. Check all the lights every now and then. Headlight aim should be checked by a technician, because a wrong tilt dazzles other drivers.

> ⚠️ **For safety and compliance:** Don’t fit brighter bulbs or “LED” bulbs inside a headlight designed for halogen unless you are sure it is legal and suitable. The wrong lighting dazzles oncoming drivers and is usually against the rules.

🚦 Warning lights: your car talks

When you start the car, most lights come on for a moment and go out; that is a self-test. The rule is simple: **red = stop safely**, and **amber (orange) = check soon**. Whoever ignores them pays more.



Simplified drawings. Warning-light shapes differ slightly between manufacturers, so check your car's manual

❌ **Don't switch the light off by disconnecting the battery or clearing the codes and carrying on.** The light is a message, not the problem. Read the code with a scan tool (OBD) and treat the cause. And any "tune" that hides faults is cheating yourself.

🚨 **Important:** If the check-engine light is **flashing**, there is probably a misfire (Misfire) that harms the catalytic converter and the engine. Ease off the load, stop safely and ask for help. A red oil-pressure light means stop now, and don't carry on "until I get home".

🎯 Challenge 3 **The red oil-pressure light comes on while you are driving. What do you do?**

A) Carry on home because it is close

B) Stop safely and switch off the engine

C) Switch it off by disconnecting the battery

🔧 What “scheduled maintenance” means, and your modifications

Minor service
Oil and filter, and a general check of fluids, tires, brakes and lights.

Major service
All of the above plus filters, spark plugs, fluids and belts as the manual says.

Full inspection
Scan-tool diagnostics and a check of suspension and steering, without necessarily changing anything.

Service history
Your invoices and the workshop stamp. It builds confidence and raises the car’s value when you sell.

> 💡 “Preventive” maintenance (before a fault) is usually cheaper than “corrective” (after a fault). It is the difference between replacing a simple filter and repairing an engine.

Maintenance after modifications: why does it need more attention? 🔥

Any modification changes the load on other parts. This is a general idea, not a recipe:

Shorter intervals: Parts such as oil, brakes and filters wear faster if you add more heat and pressure.

Fluids that suit the use: They may need a higher specification. Ask the part maker or a specialist technician, and don’t guess.

Earlier checks: Leaks, noises, heat, bushings. A modified car “talks” sooner.

Remember and record: Write down every change. And “premium oil” doesn’t cancel the intervals, and no modification exempts you from the manual.

After a repair: a simple “run-in” 🌱

A new part needs a settling-in period, without pushing it:

New pads and discs
Drive gently for the first days and avoid sudden hard braking until the surfaces bed in. Follow the manufacturer’s instructions.

New tires
Be careful for the first days. A new surface has slightly less grip before it roughens.

After changing a wheel or rim
Ask for the nuts to be re-tightened after a short distance, to a specified torque (a torque wrench).

After an engine repair
Follow the workshop’s instructions exactly: speeds, revs and an earlier oil check.

Storing the car 🅿️

Clean and dry it, and keep it in shade or under cover. The sun is hard on the interior and paint.

Fresh oil and enough fuel as the manufacturer advises. A car stored for a long time needs a fuel stabiliser or a technician’s advice.

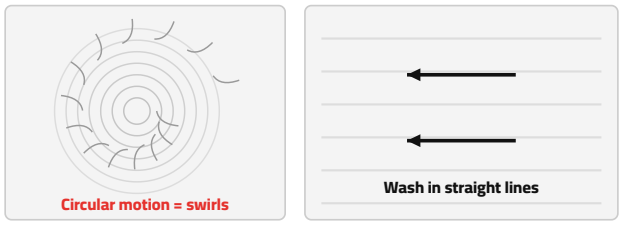
A maintenance charger for the battery, or disconnect it as the manual says.

Tire pressure as the manufacturer advises for long storage. And don’t leave the handbrake on for a long time without need.

Actually move it and drive it every so often, not just start it in the parking spot.

✨ Paint and interior care

Washing the right way 🧼



Illustrative drawing: circular motion leaves swirls (Swirl Marks), straight motion shows less

Wash in the shade and on cool paint, because water and soap dry fast and leave marks.

Two buckets: one for soap and one for rinsing the mitt. And car shampoo, not dish soap, which strips the wax.

Don't wipe dust when it is dry! Gulf dust is tiny grains that scratch. Rinse with water first.

A clean microfibre cloth, and dry with a soft towel.

Wax, sealant, ceramic, paint protection: an honest comparison 🧪

Type	What is it	The truth, without exaggeration
Wax	Usually a natural layer (carnauba)	A warm shine, but it needs renewing fairly often (weeks to a few months)
Sealant	Synthetic polymer	Usually longer protection than wax; renew every few months
Ceramic	A silica coating	A harder, longer-lasting layer, but not a "scratch shield", and it needs washing and care
Paint protection film (PPF)	Clear polyurethane film	Absorbs stone chips and light scratches. Higher cost, and it should be fitted by an expert

> 💡 Good preparation decides the result more than the product name. And never believe any "never scratches" promise.

Interior, lighting and sun 🌞

Interior
Vacuum and a cleaner suited to the material. Leather needs conditioner. And never polish the steering wheel or pedals with slippery products! 🚫

Sun
A windscreen sunshade, and shade whenever possible. Clean and moisturise the rubber with suitable products. Window tint within the rules.

Headlights and dust
Plastic yellows from UV; polishing and a protective layer from a specialist help. And Gulf dust deserves more frequent air and cabin filter changes.

🎯 Challenge 4 **What most often causes swirl marks in paint?**

A) Washing with one bucket and a dirty cloth | B) Cold water | C) Wax

🕵️ Buying a used car: the detective's checklist

Before you fall for the car, inspect it. Falling in love is the most expensive way to buy. 😄

1. **Papers:** The serial number (VIN) matches the papers, and the ownership, inspection and insurance are valid. Ask for the service history.
2. **Paint and body:** Differences between colours, or in the gaps between doors and bonnet, or paint overspray on rubber. Signs of a past accident.
3. **Tires:** All four the same type? Uneven wear? Age?
4. **Under the bonnet:** Burnt black oil, or milky residue under the cap, leaks, neat or messy wiring.
5. **Cold start:** Any strange noise or smoke at start-up? Ask them not to start it before you arrive.
6. **The test drive:** Is the steering straight? Do the brakes shudder? Does the gearbox clunk? Is the A/C cold? Do all the buttons work?
7. **Warning lights:** They should come on and then go out at start-up. A light that never comes on may mean it was **disconnected on purpose**! 🤨
8. **Independent inspection:** A technician or inspection centre of your choosing, not the seller's choice.

> 💡 Ask frankly: is the car modified? Are the original parts still there? A legally modified car with invoices and professional fitting can be excellent, but an "undocumented modified" car needs double the caution.

How do you spot modification damage? 🔎

Signs of hard use

Abnormal wear on tires and brakes, a smell of burnt clutch, or an engine oil leak.

Unprofessional work

Haphazardly wrapped wiring, rough welds, missing bolts, or mismatched parts.

Unknown tuning

Ask about the tune, its software and invoices. A scan tool sometimes reveals changes or stored codes.

Removed legal parts

Any removal of emissions or safety systems is against the rules, and it becomes your problem at inspection and registration. Walk away.

> ❌ Buying a "beautiful" car at an excellent price without an independent inspection and without records. Often the beauty hides the story.

🤝 How do you choose a workshop you trust?

The best technician is the one who explains the reason, not the one who gives you a number with no explanation. Questions to ask before you hand over the key:

"What exactly is the diagnosis?"
The technician explains the fault and the evidence, not "let's replace and see". Replacing by trial costs you.

"A written quote?"
Before the work, with parts and labour itemised. And any extra needs your permission.

"What brand is the part?"
The itemised invoice has the part name, number and price. And ask for the old parts back.

"Is there a warranty?"
A warranty on the work and the part, written on the invoice, with a clear period.

Genuine or aftermarket parts? 🔩

Type	What is it	Note
Genuine (OEM)	From the car's manufacturer	Usually the most trusted, and usually the most expensive
Same maker, different box	The same factory, different packaging	Original quality at a lower price, sometimes
Replacement (Aftermarket)	Another independent company	Quality varies from excellent to poor, so choose a known brand
Counterfeit	Copied with fake packaging	Dangerous! Buy only from a trusted source

✅ **DE.AUTO tip:** Keep every invoice. It is your car's "CV", and it matters when you sell or if a dispute arises.

❌ Choosing a workshop only by the cheapest price. The cheap one that redoes the job twice costs far more.

🗣️ Garage talk vs the proper name

Gulf garage talk	Proper name	What it means
"Sawwi li tcheek"	Check-up (Inspection)	Inspection and diagnosis
"Ghayyir li ad-dihn"	Oil change	Engine oil with the filter
"Mayy ar-raditer"	Coolant	Water with antifreeze and anti-rust
"Zayt faramil"	Brake Fluid	Hydraulic fluid for the brakes
"Ad-dinamo"	Alternator	Charges the battery
"Sayr al-krank"	Accessory / timing belt	Ask which belt they mean
"Mizan wa tarsees"	Alignment + wheel balancing	Chapter 10
"Kombiyutar as-sayyara"	ECU	The engine control unit

🧨 Maintenance myths and mistakes

✕ MYTH	✓ FACT
"Warm the engine for ten minutes before you drive off."	In modern cars a few seconds is usually enough, then drive gently until it reaches operating temperature. Long idling wastes fuel and increases deposits.
✕ MYTH	✓ FACT
"Expensive premium oil means I never change it."	Any oil is affected by heat, fuel and moisture. The filter saturates, and your specification follows the manual. "Premium" gives better protection, but it doesn't cancel the interval.
✕ MYTH	✓ FACT
"If the engine light came on and went off, there is nothing wrong."	A light that switches itself off may have a stored code behind it. Read it with a scan tool before it grows.
✕ MYTH	✓ FACT
"More oil is better than too little."	Overfilling above MAX can make the oil foam and harm the engine. The correct level is best.

Common mistakes ❌

- ❌ Opening the radiator cap while the engine is hot.
- ❌ Mixing types of coolant or brake fluid without checking the specification.
- ❌ Ignoring a warning light "until we get free time".
- ❌ Washing the engine bay with a strong water jet over wiring and sensors.
- ❌ Tightening wheel nuts "by eye" without a set torque.
- ❌ Postponing maintenance because the car "runs fine". Faults start silently.
- ❌ Modifying the car and neglecting brakes, tires and fluids. Power without maintenance = danger.

🎯 Challenge 5 **What is the most suitable approach to fluids and intervals if your car is modified?**

A) Drop the manual and follow my friend

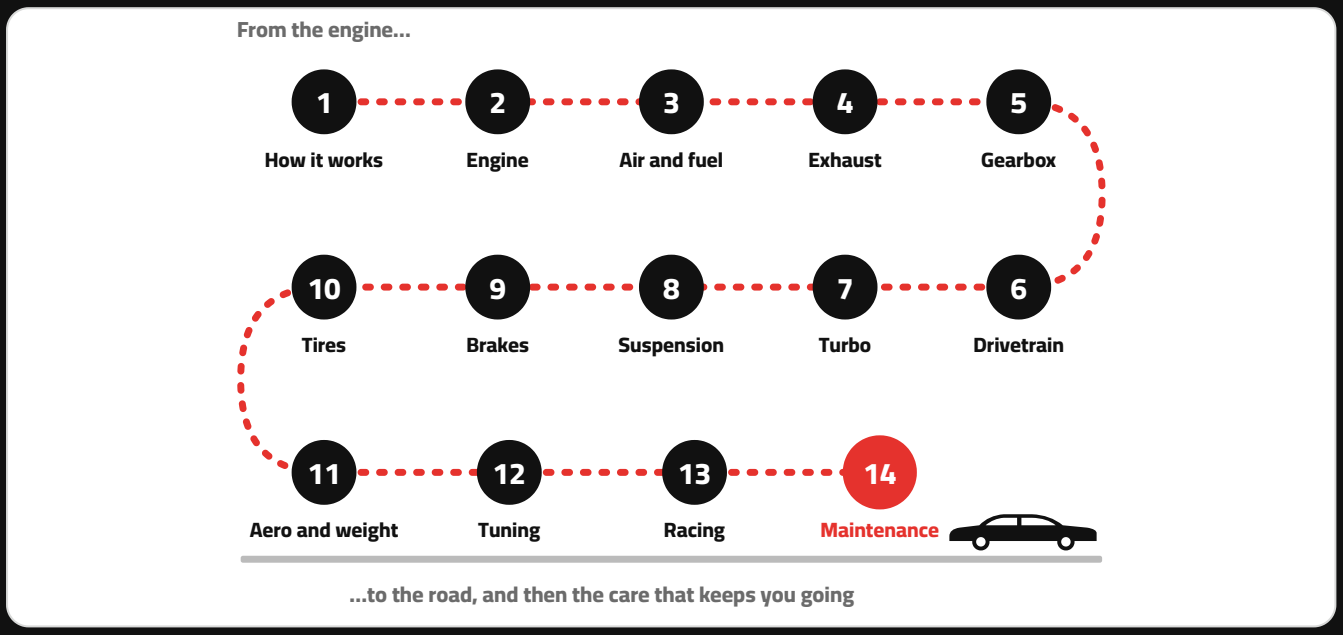
B) Shorten intervals, check earlier and consult a specialist

C) Nothing changes

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The finish: we've reached the last line 🏁

14 chapters, and each one was a stop on a single journey



The fourteen chapters: power is born in the engine and passes through every system until it reaches the ground

From the first moment we asked "how does a car move?" until we reached the last thing that keeps it moving: **care**. The power generated in the engine, passing through the gearbox and differential and reaching the wheels, means nothing if your car isn't healthy. And every modification, however nice, starts from a sound base.

Now when you hear an exhaust note, or read a tire size, or see a strange word in the workshop, you know what is behind it, and you can chat with car people comfortably. And most important of all: you know how to look after your car, and to choose safety before showing off, and the rules before the challenge.

Thank you for being with us through every chapter. A good car is one that is well looked after. See you on the road, and safe travels always. 🚗💨

🎯 Challenge answers:

1) Fluidity when cold. 2) Open it after it has cooled down completely. 3) Stop safely and switch off the engine. 4) Washing with one bucket and a dirty cloth. 5) Shorten intervals, check earlier and consult a specialist.

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